

Design of LC Voltage Controlled Oscillator Using Cross-Coupled MOSFET Pair

EE 299

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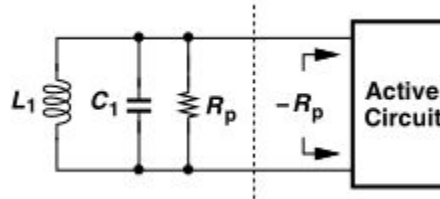
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RLC circuit

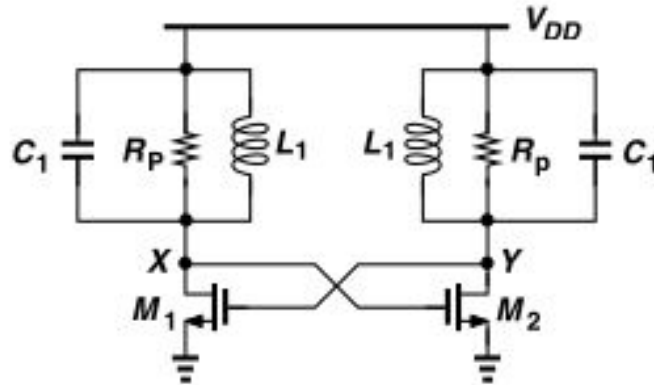
Resistance (The Lossy Tank)

- Real-world oscillators incorporate an inductor which has a physical series resistance (R_s).
- This lossy inductor is modeled as an equivalent parallel resistance (R_p) at resonance.
- This resistance R_p drains energy from the LC tank, causing the oscillation amplitude to decay over time rather than being self-sustaining.
- To keep the signal alive, we need a circuit that acts like a negative resistor to cancel out that positive resistance, which would add energy which is being drained by the resistor.
- By providing this energy, the circuit allows its own internal noise to grow until it becomes a stable, periodic signal.



Cross-Coupled Pair: Analysis

- **Purpose:** One of the most common ways to generate the required negative resistance is using a cross-coupled differential pair.
- **Basic Topology:** Two transistors (M_1 and M_2) are connected such that the gate of M_1 is tied to the drain of M_2 , and the gate of M_2 is tied to the drain of M_1 .

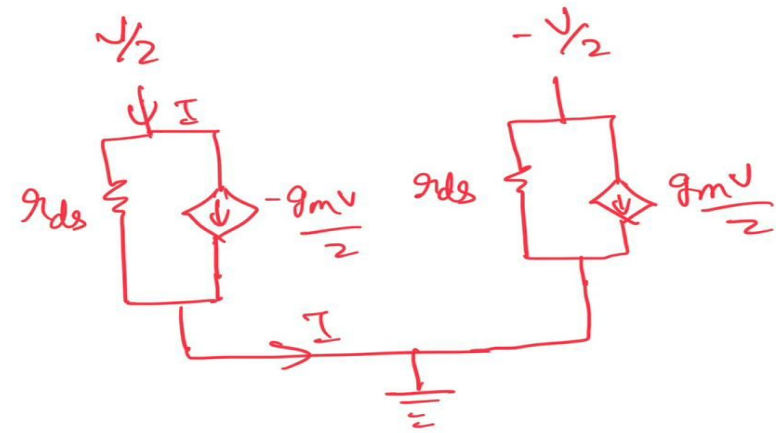


- **Small-Signal Analysis:**
 - If we apply a differential voltage v_x across the two drains, the gate-source voltage of one transistor increases while the other decreases.
 - This causes a current i_x to flow **out** of the terminal where voltage is applied, which is the definition of negative resistance.
- **Calculated Resistance:** The equivalent resistance looking into the drains of the cross-coupled pair is:

$$R_{in} = -2/g_m$$

- **Startup Condition:** To ensure the circuit starts oscillating from noise, the cross-coupled pair must be "strong" enough to overcome the tank losses:

$$g_m R_p \geq 2$$

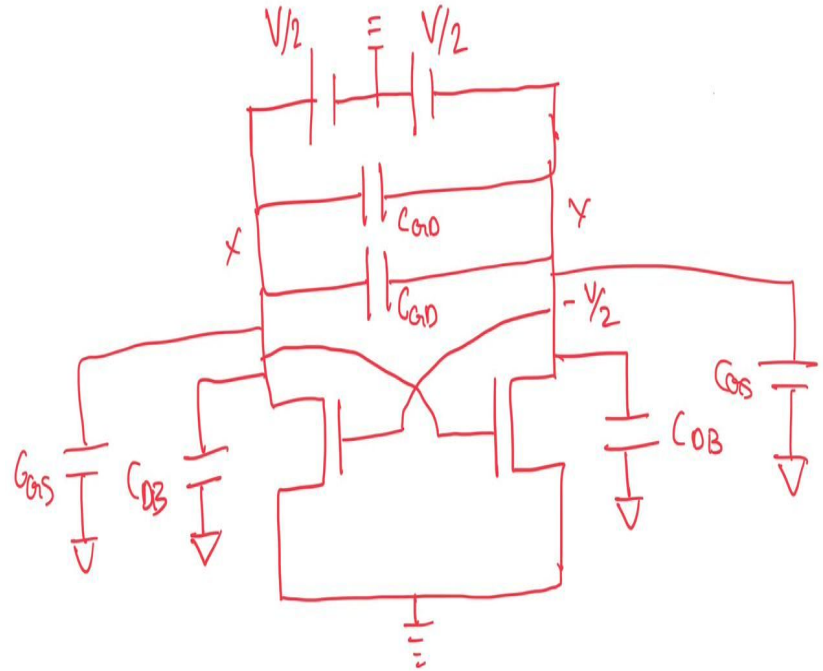


$$\frac{V}{2} r_{ds} + \left(-\frac{gmV}{2} \right) = I$$

$$\frac{V}{I} = \frac{2}{-gm + \frac{1}{r_{ds}}}$$

$$r_{ds} \gg gm$$

$$R_p \approx -\frac{2}{gm}$$



$$C_t = 2C_{GD} + \frac{C_{GS}}{2} + \frac{C_{GB}}{2}$$

Why Add a Tail Current Source (ISS)?

- **Bias Stability:** Defines a fixed total current for the cross-coupled pair, making the circuit's operation independent of small variations in the supply voltage (VDD).
- **Amplitude Control:** Directly sets the differential output swing; since $V_{\text{peak}} \approx I_{\text{SS}} \cdot R_p$, you can precisely tune the signal strength.
- **Power Management:** Acts as a current limiter to prevent the transistors from drawing excessive power, which would occur in a "tail-less" configuration.

Cross - Coupled LC Oscillator

$$L = 340 \text{ p H}$$

$$C = 15 \text{ p} + (\text{C from capacitance bank}) + (\text{C from Varactor}) + (\text{Parasitic Capacitance from cross-coupled network})$$

$$R_p = QWL$$

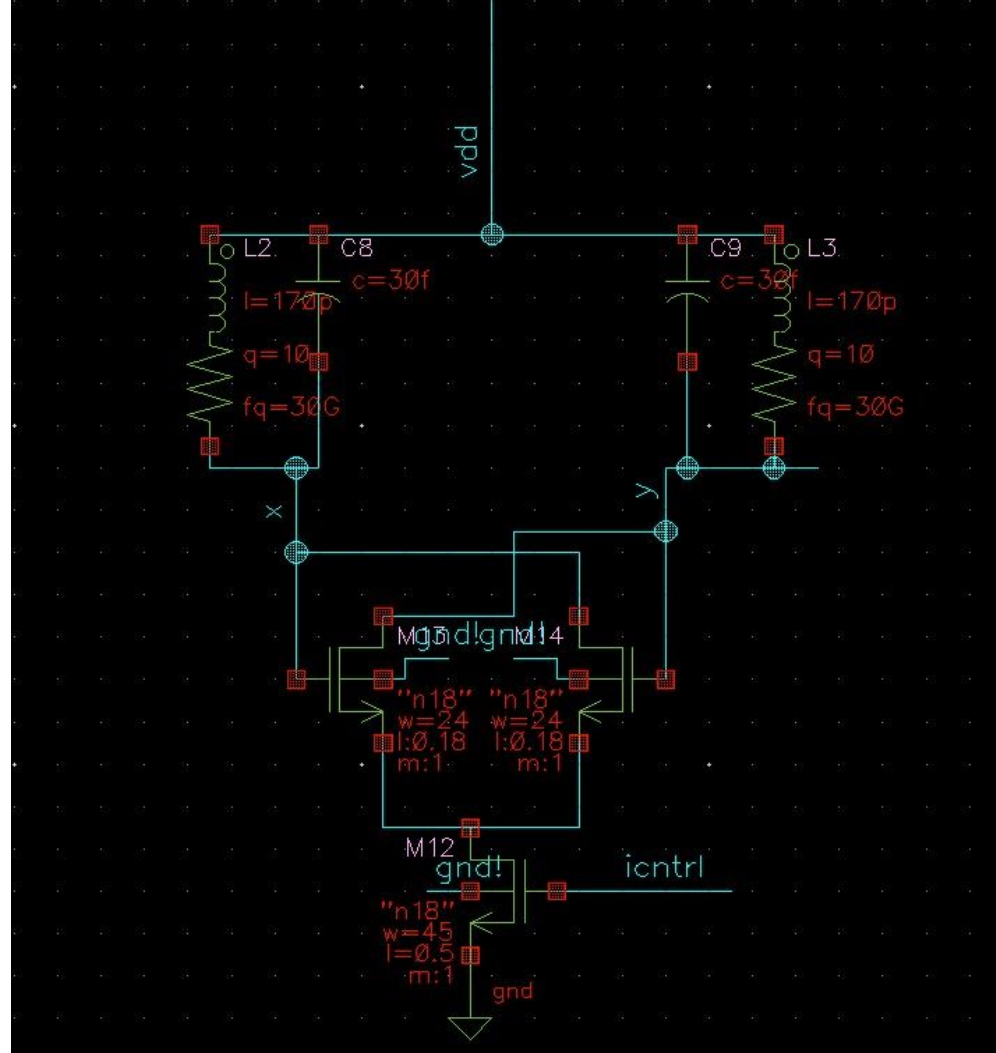
$$R_p @ 27.05 \text{ G Hz} = 627.69 \text{ ohms}$$

$$R_p @ 32.1 \text{ G Hz} = 746.253 \text{ ohms}$$

The corresponding gm for oscillations are:

$$g_m @ 27.05 \text{ G Hz} > 3.18 \text{ m}$$

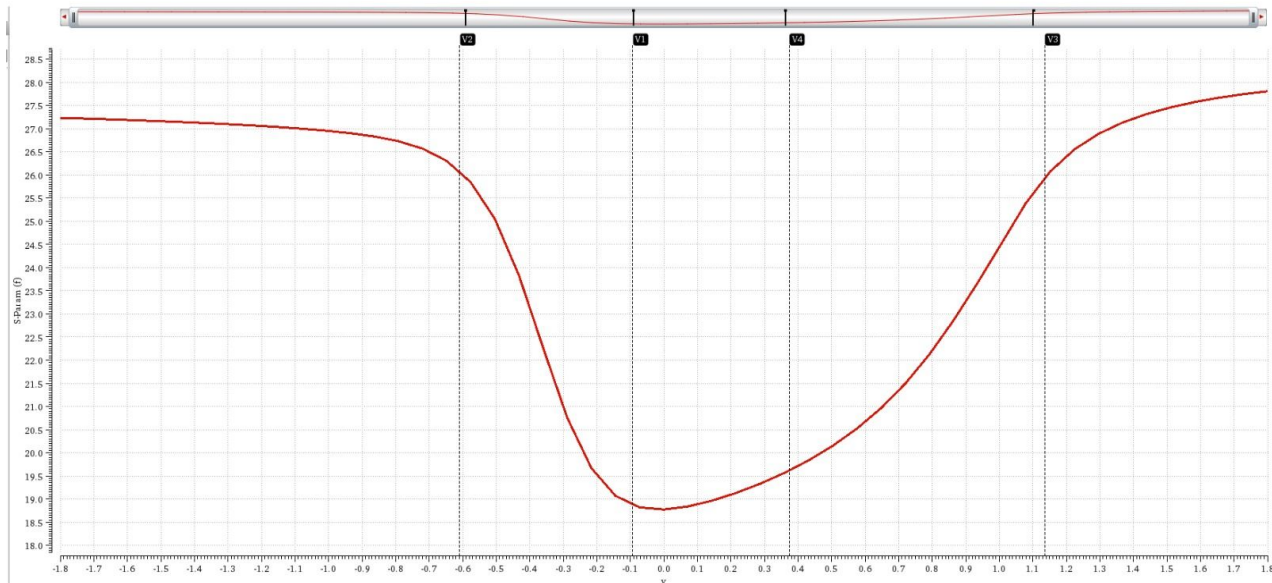
$$g_m @ 32.1 \text{ G Hz} > 2.68 \text{ m}$$



A single MOS varactor has a very non-linear curve. It changes capacitance quickly in the middle but very slowly at the ends. This makes the oscillator's behavior inconsistent across different frequencies.

The range of the capacitance change is not that huge, difficult to get a wide tuning.

Increasing the size of the mosfet could solve the problem, however adds parasitic capacitance, which do not allow us to hit the desired frequencies.

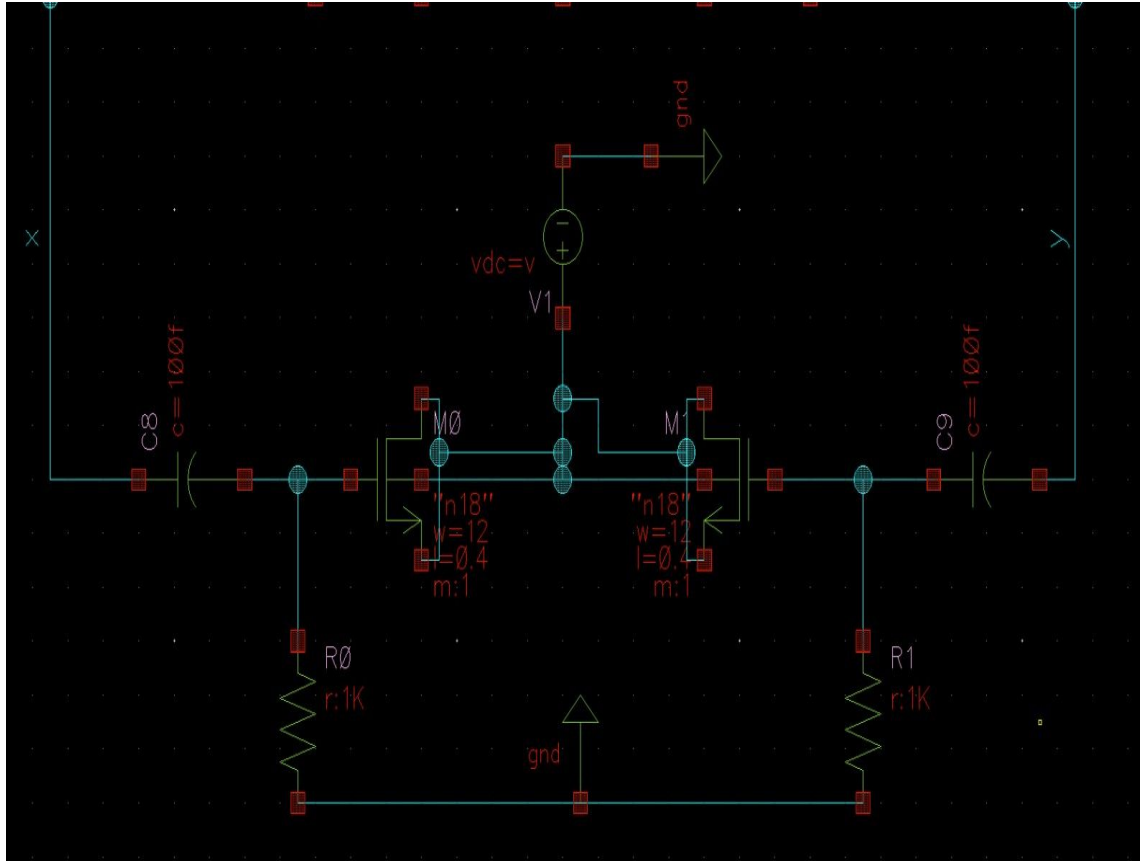


Switched Capacitor Banks

Discrete Tuning : Uses a bank of fixed capacitors connected to the LC tank through MOSFET switches. These switches are controlled by digital signals.

This technique helps as a single varactor isn't force to cover wide ranges .

Fine Tuning : Uses a small MOS Varactor whose capacitance changes continuously with an analog control voltage (V_{ctrl}).



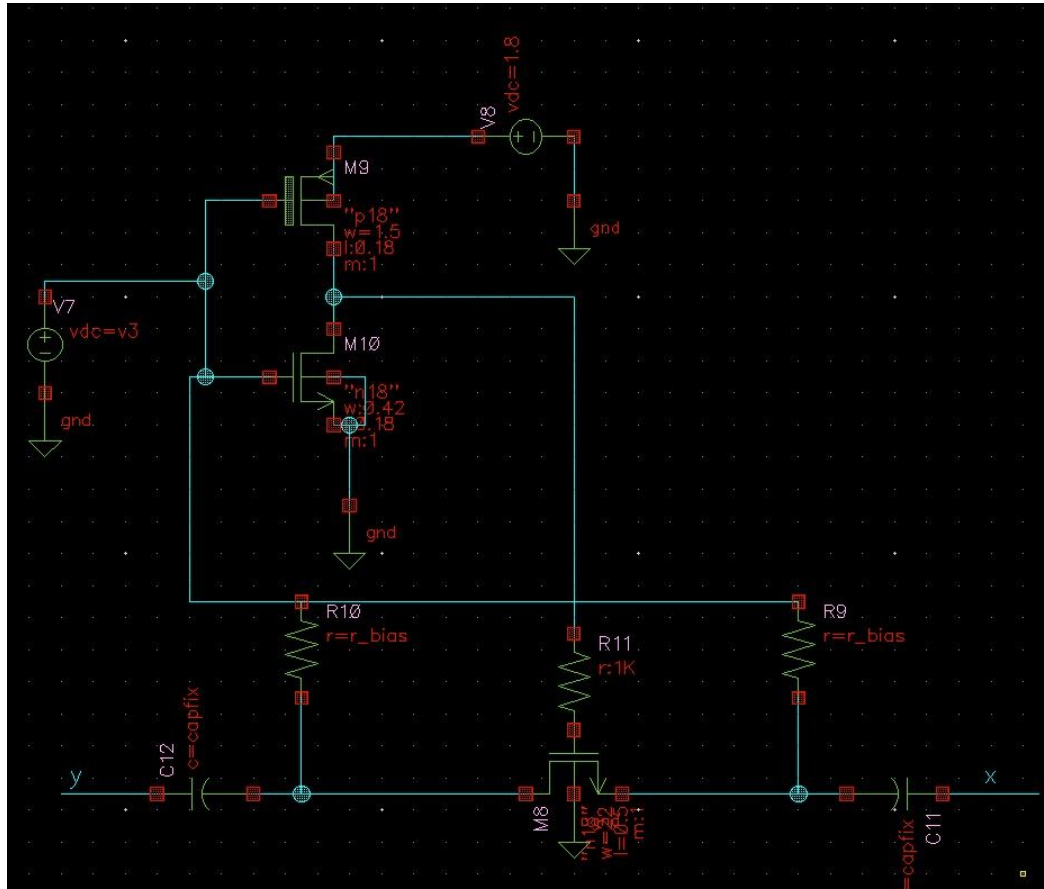
Varactor Circuit:

A capacitor is attached because it allows you to apply the Control Voltage (V_{ctrl}) directly to the varactor without affecting the bias of the cross-coupled transistors.

The capacitor acts as a DC block and allows a ac path.

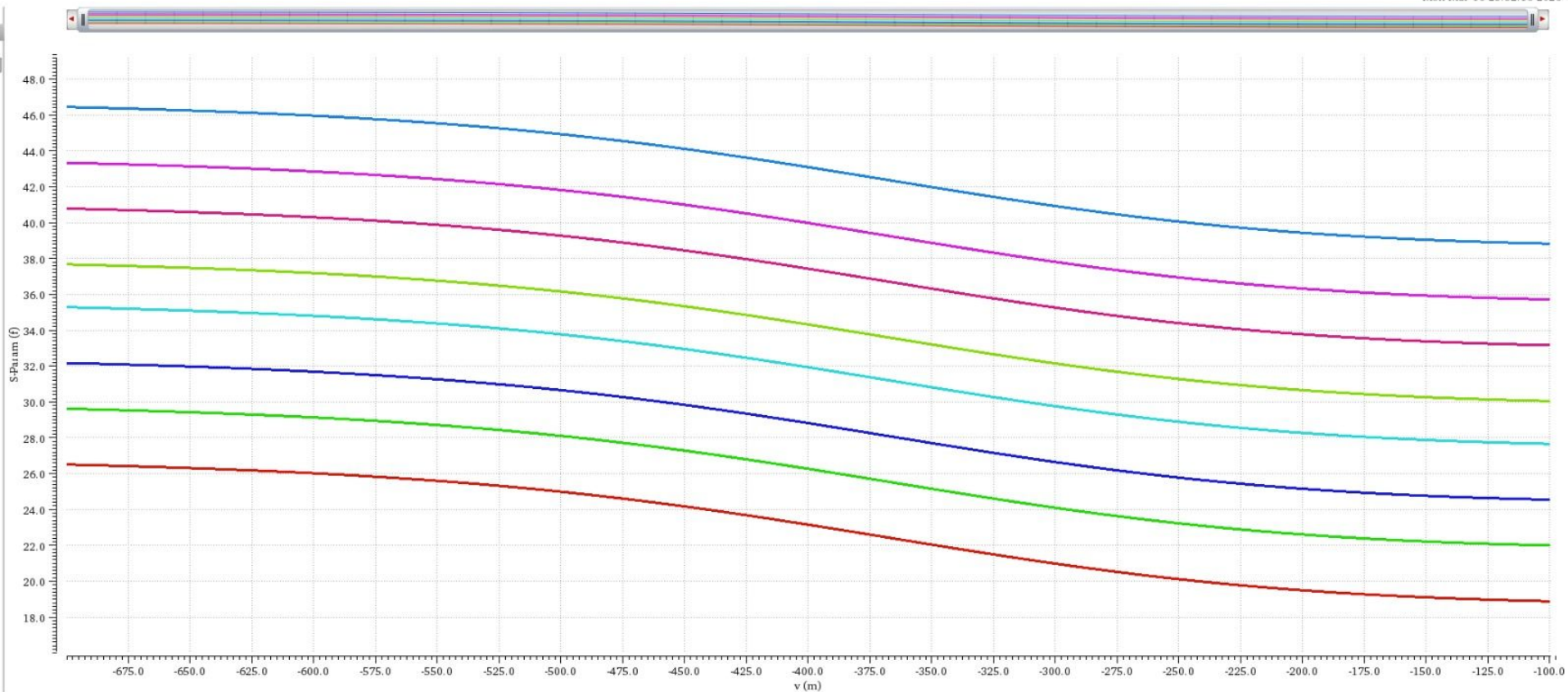
Capacitance Bank

The capacitance bank consists of 2 cap connected via nmos switch.



Capacitance Tuning

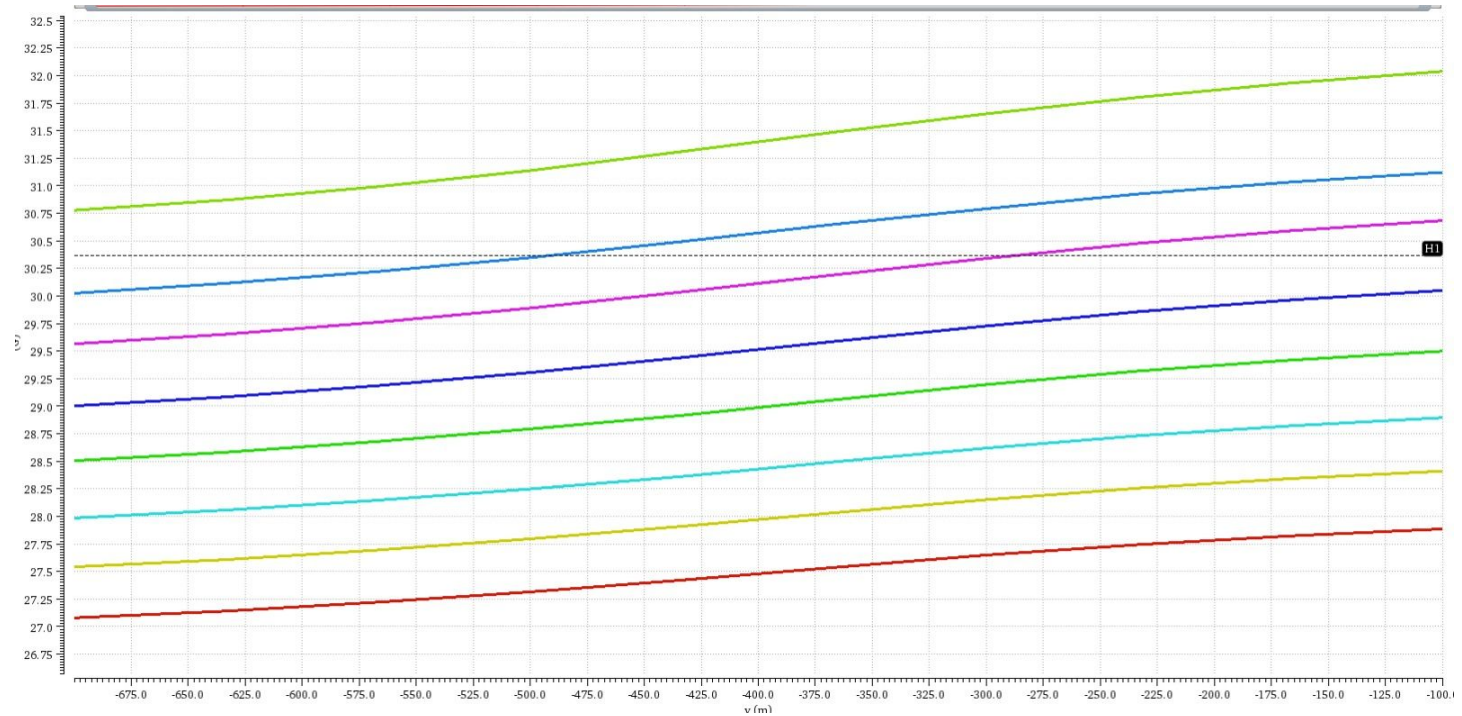
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Tuning Range : 17.07 %

Max Freq = 32.1GHz

Min Freq = 27.05 GHz



Reason for Voltage Swing Control

Fixed Tail Current (ISS): In our design, the tail current source sets the total current. The peak differential voltage swing is approximately:

$$V_{\text{peak}} \approx I_{\text{SS}} \cdot R_p$$

where R_p is the equivalent parallel resistance of the tank.

Inductor Quality Factor (Q): The resistance R_p is not a constant; it depends on the Quality Factor of the inductor (QL) and the frequency of operation (ω):

The Frequency Factor: When you increase the capacitance (C), the resonant frequency ($\omega=1/LC$) **decreases**.

The "Tank Loss" Effect: As the frequency ω drops:

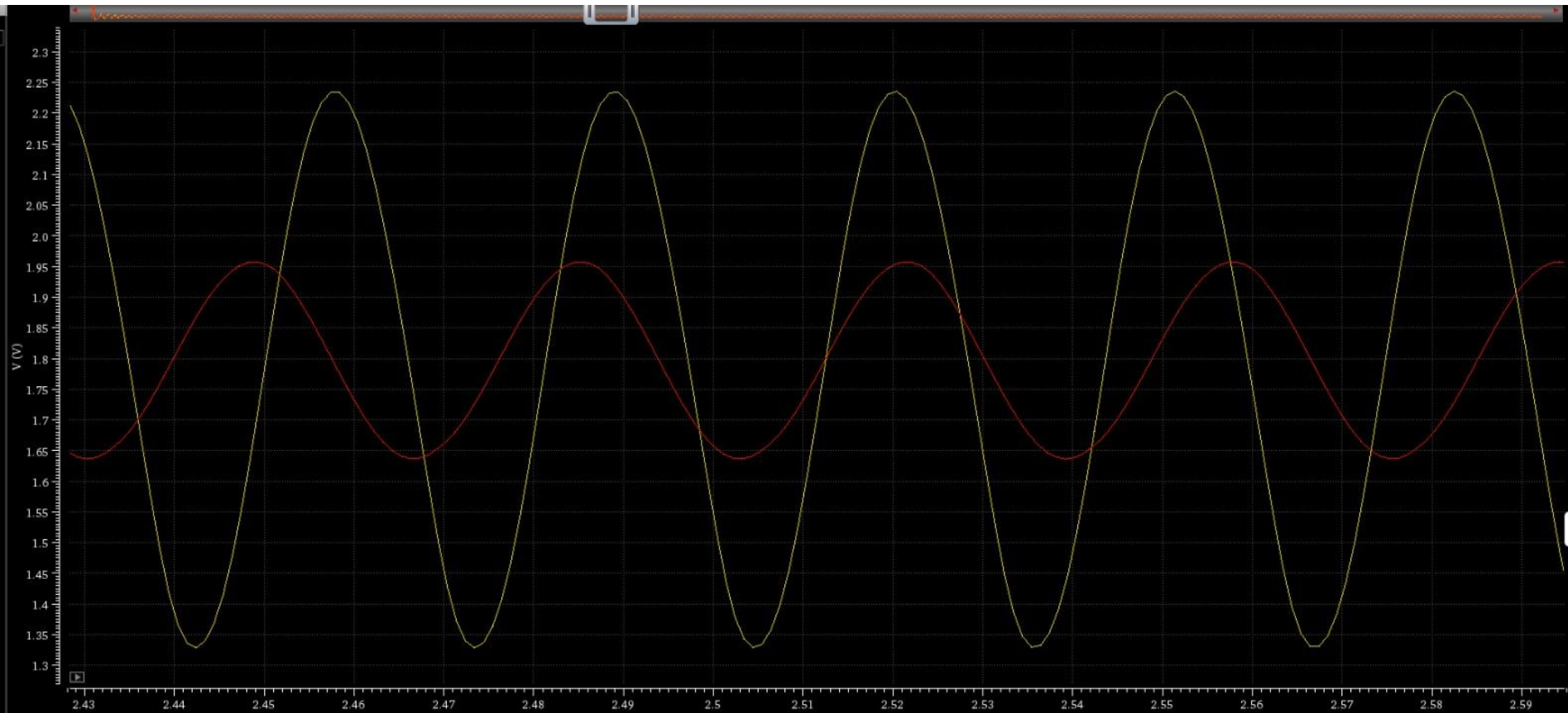
1. The term ωL decreases.
2. Consequently, the equivalent parallel resistance **R_p decreases**.
3. Since the voltage swing is $I_{\text{SS}} \times R_p$, a lower R_p results in a **smaller voltage swing**

To tackle this issue we must increase the voltage swing by increasing the tail current by:

$$Q = \frac{L_1 \omega}{R_S} = \frac{R_p}{L_1 \omega}$$

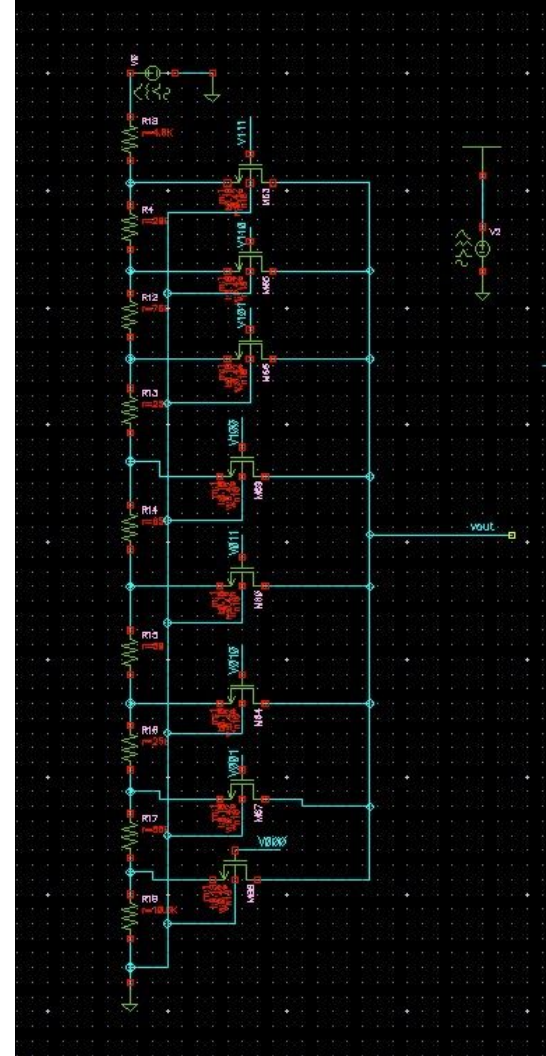
$$R_p = \frac{L_1^2 \omega^2}{R_S}$$

Reason For Voltage control:



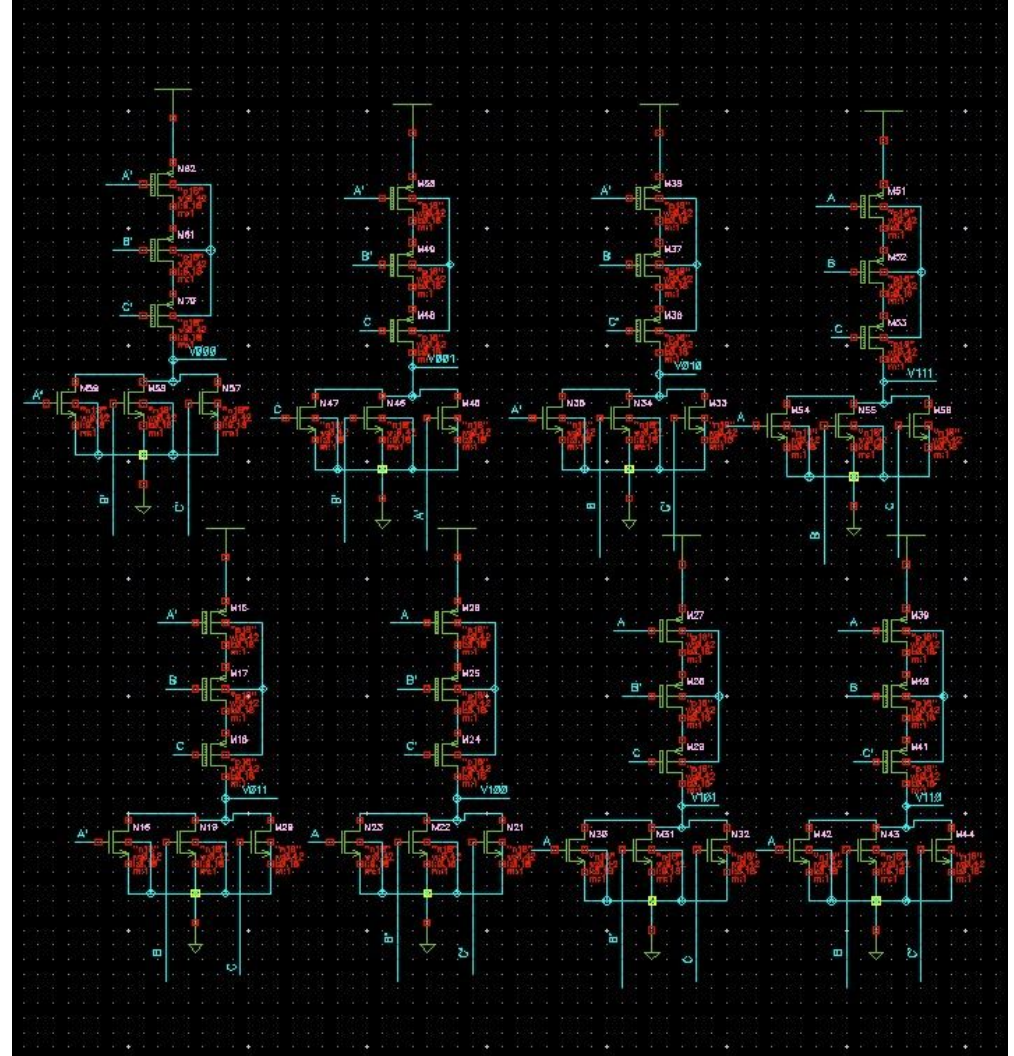
Voltage Divider

Design of a CMOS Pass gate logic voltage divider that sets the Gate voltage according to the operation of the capacitor bits.

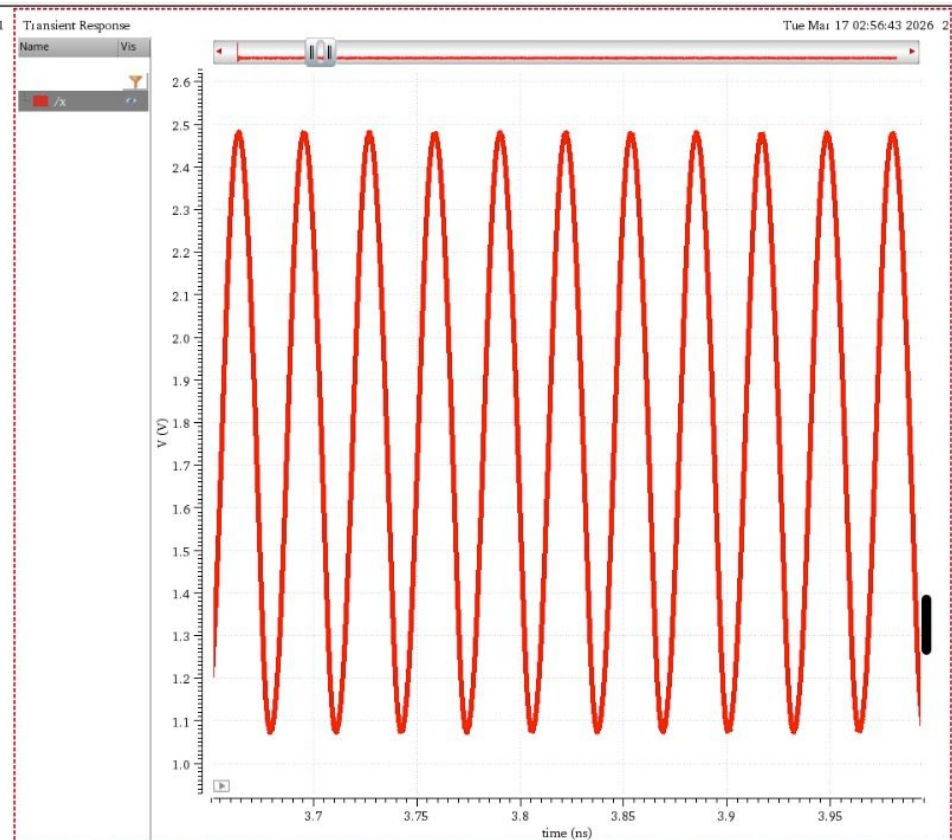
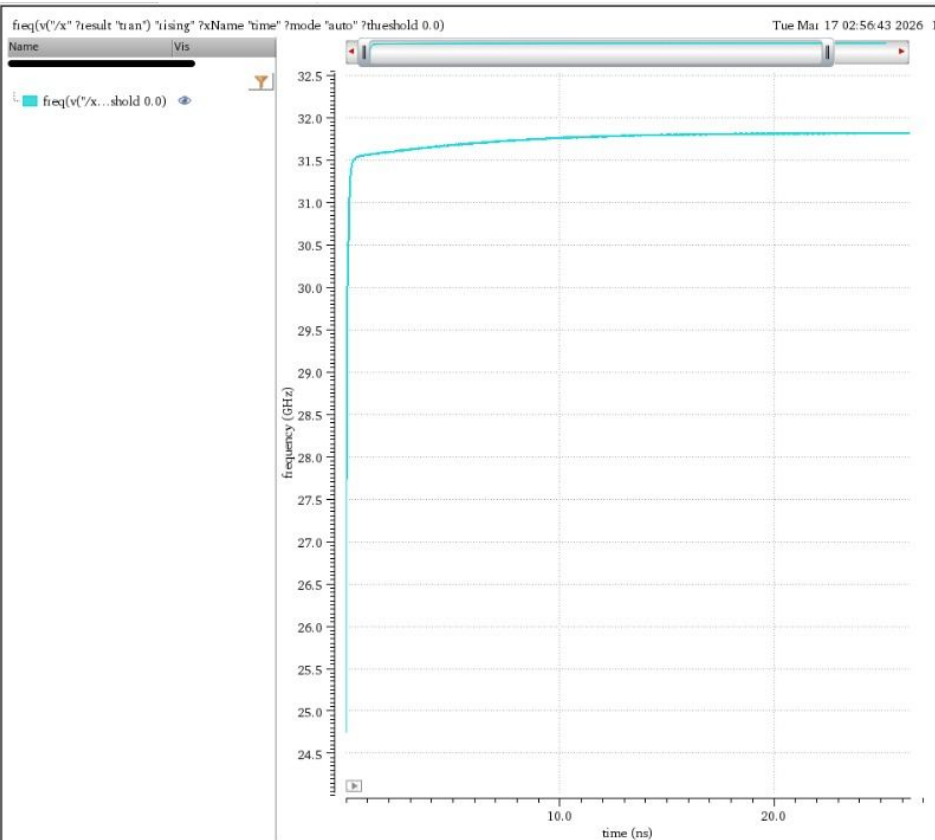


Cmos 3 input And gate

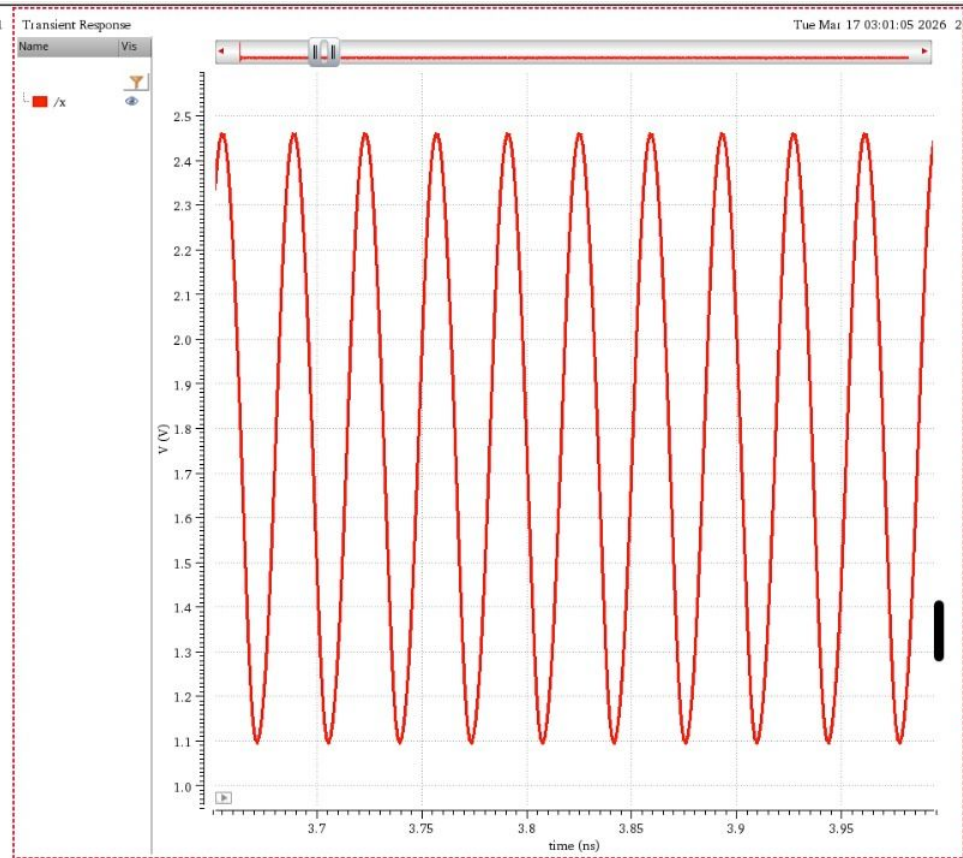
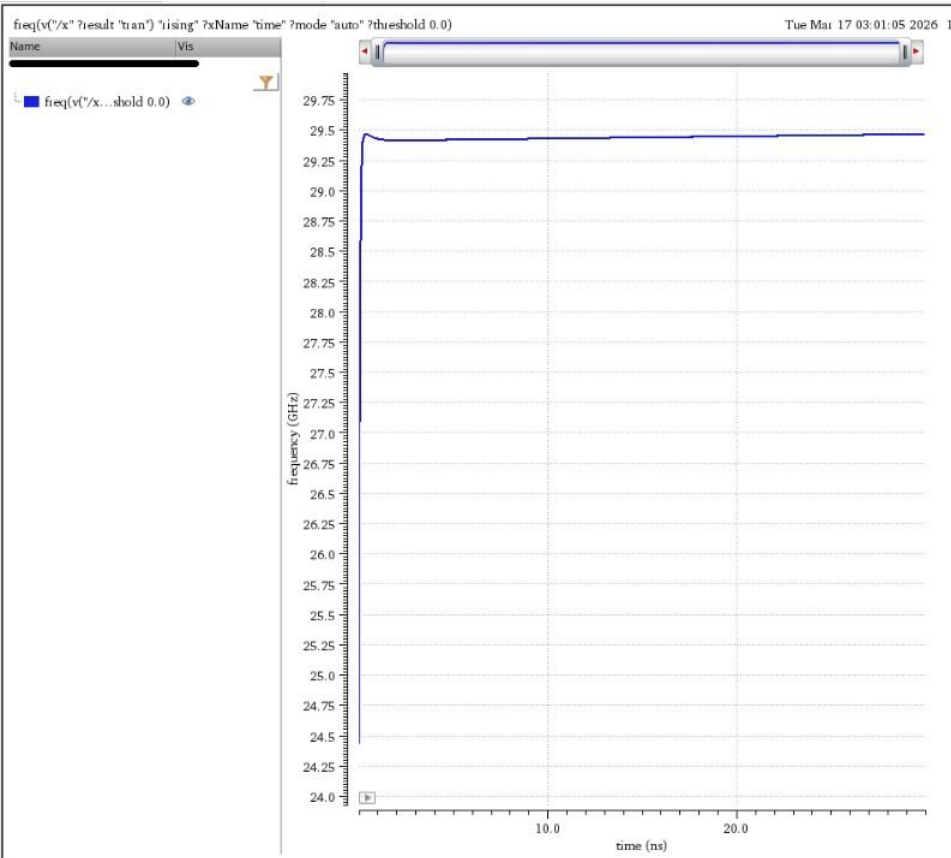
Design of 3 input AND Gate for passing different Voltage to the Gate of the tail current source to maintain the Voltage swing.



Output when Capacitor Bit(110)



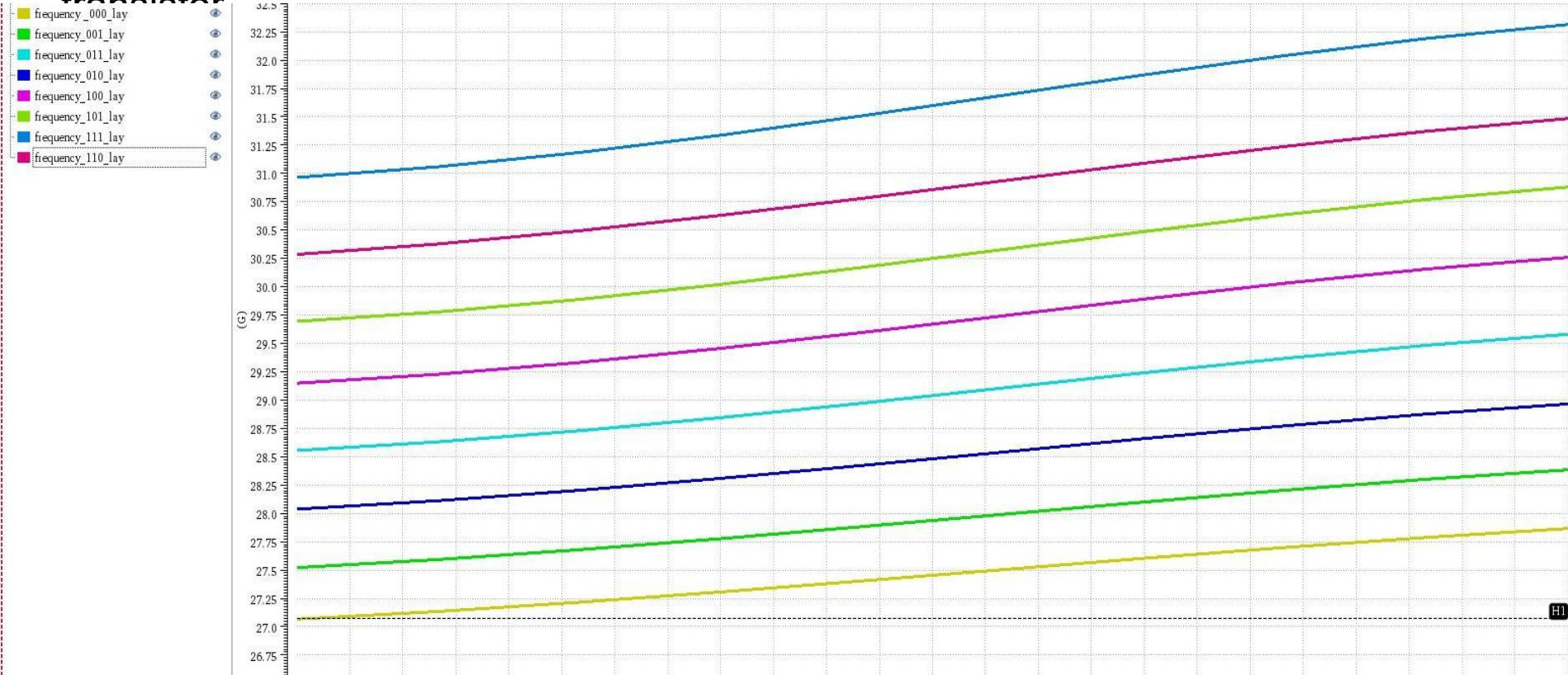
Output when Capacitor Bit(010)

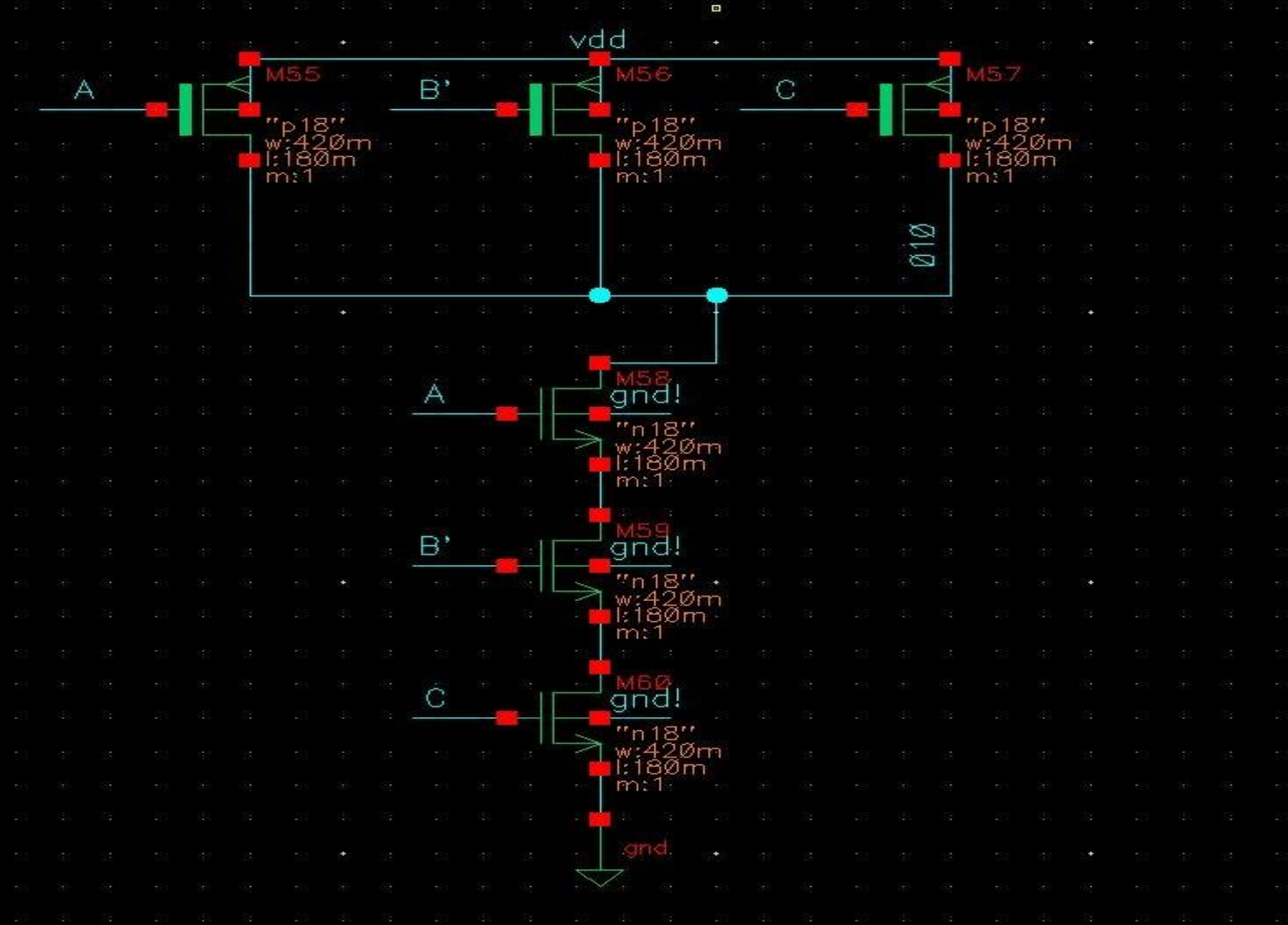


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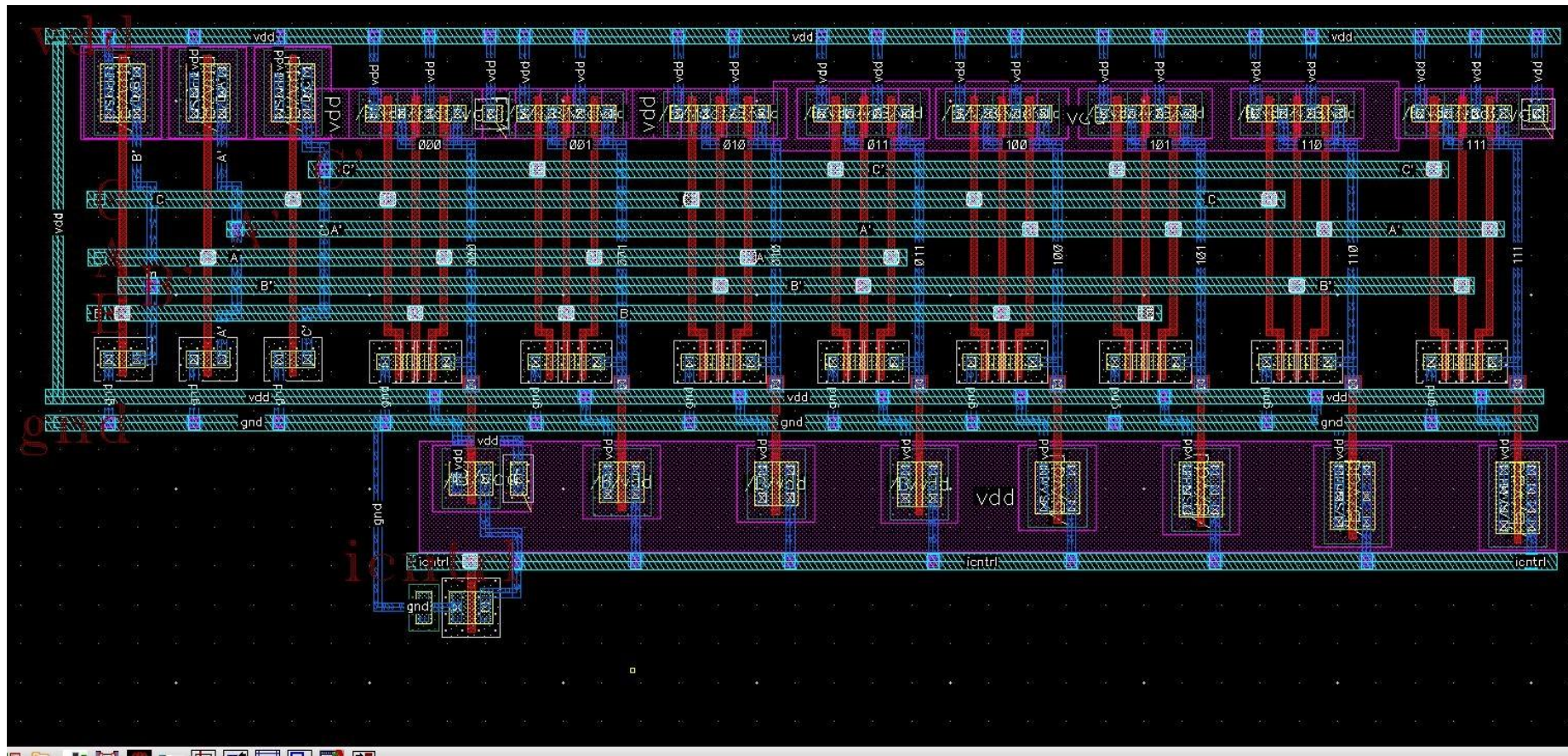
Completed the layout of VCO cross coupled and the tail

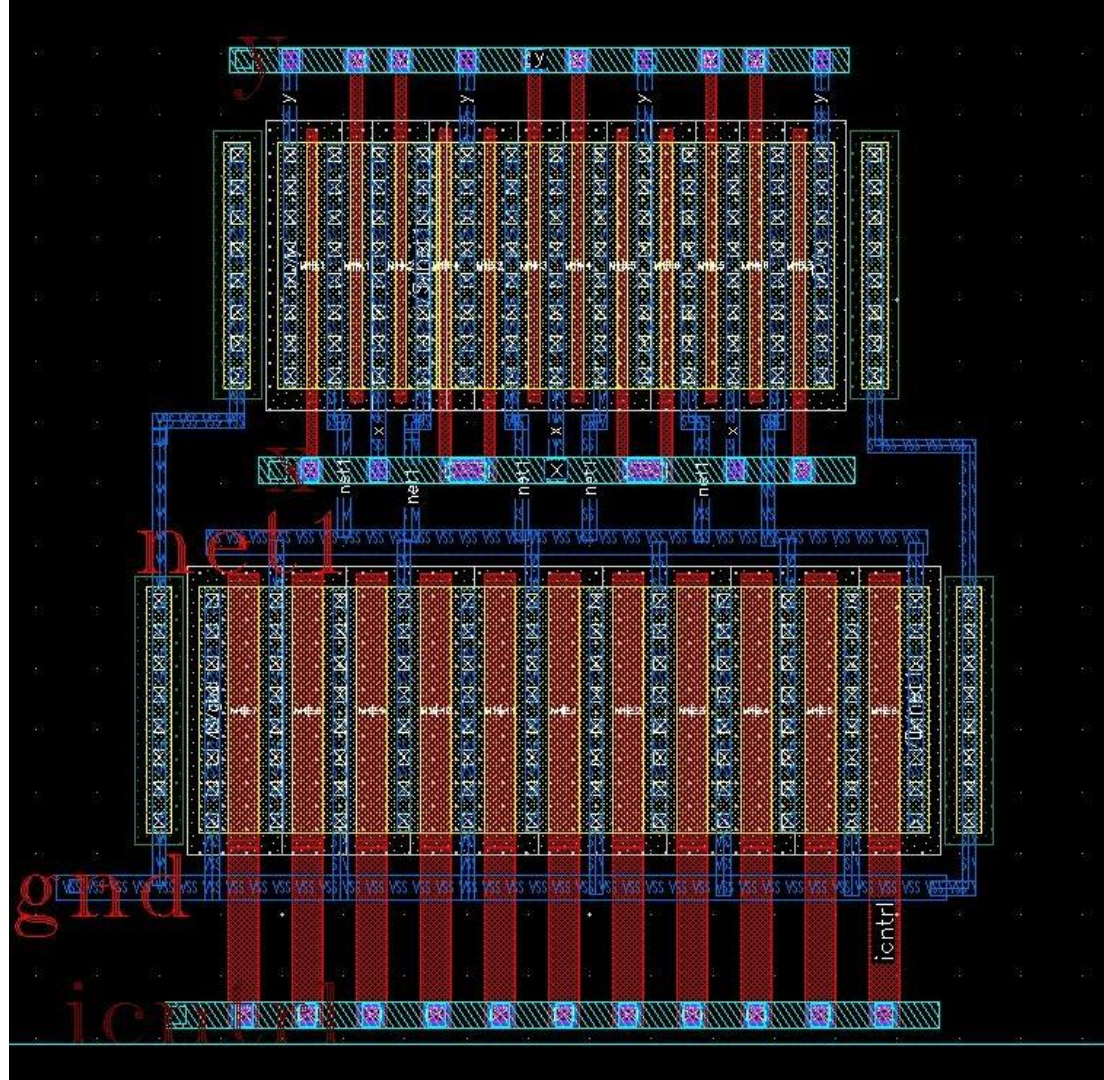
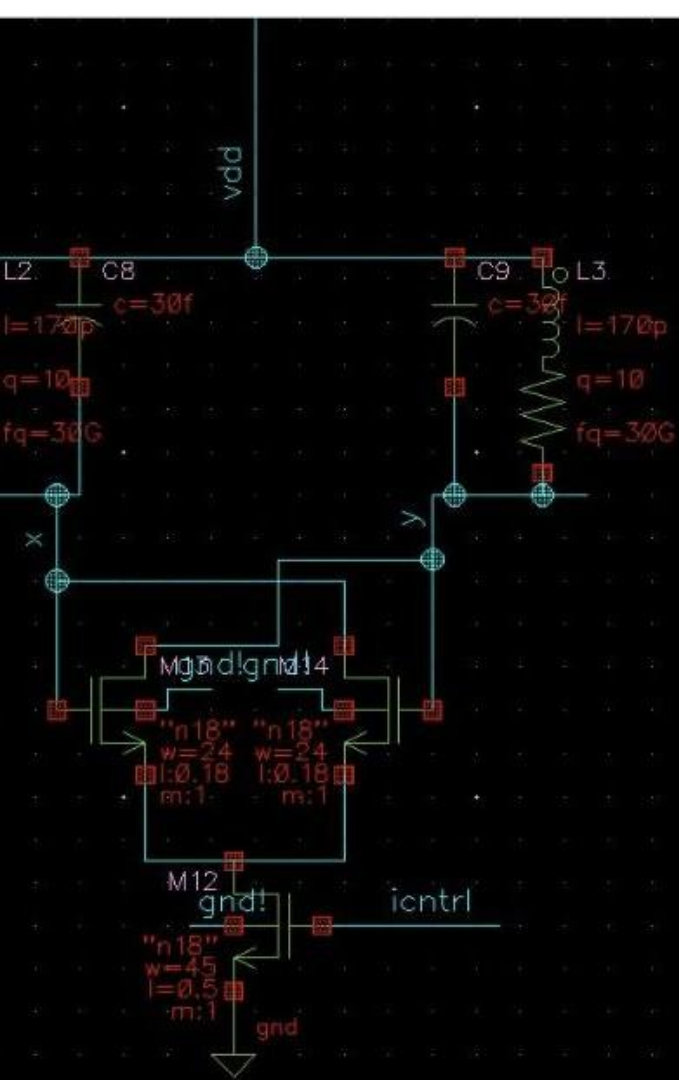
transistor



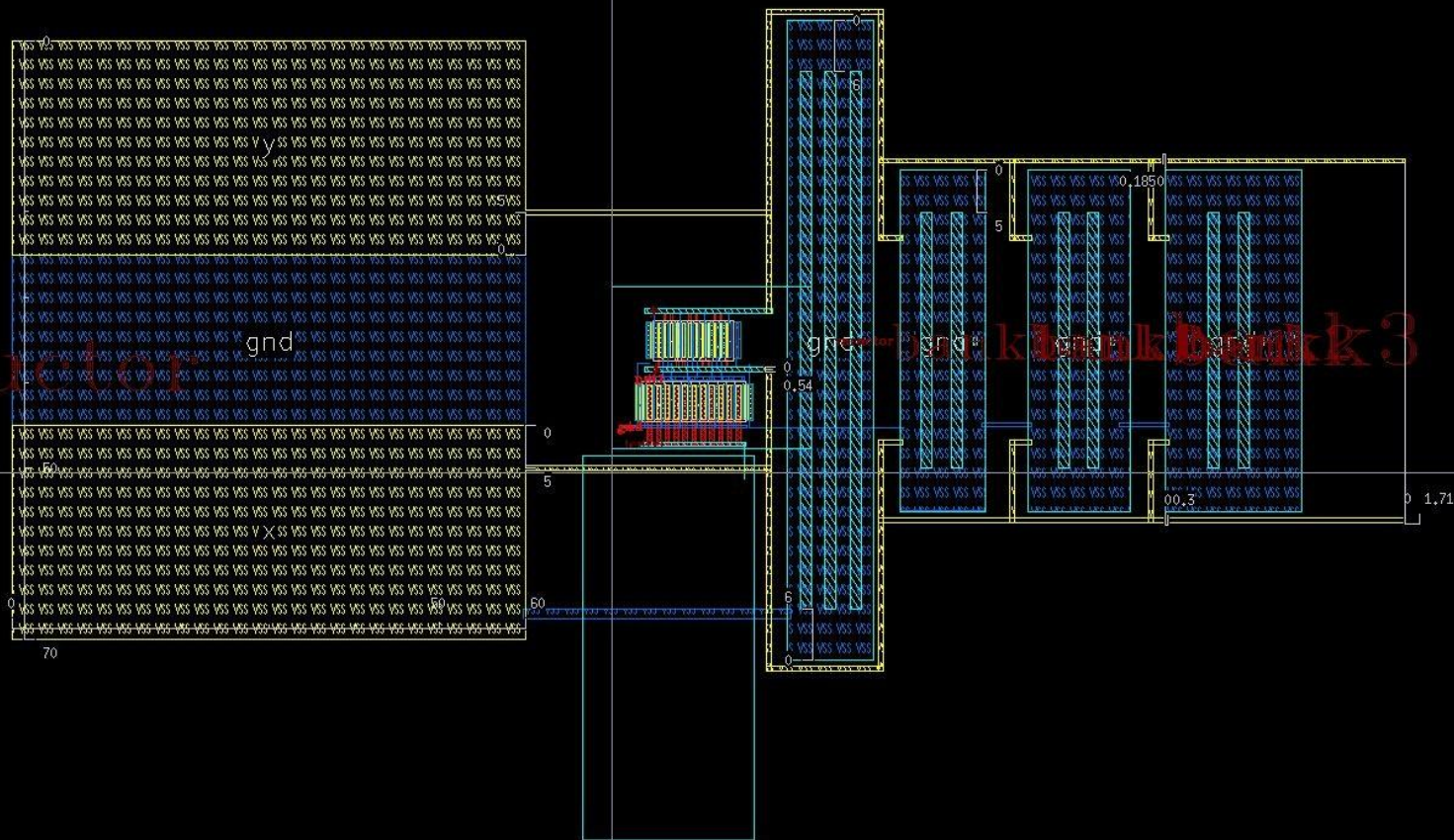


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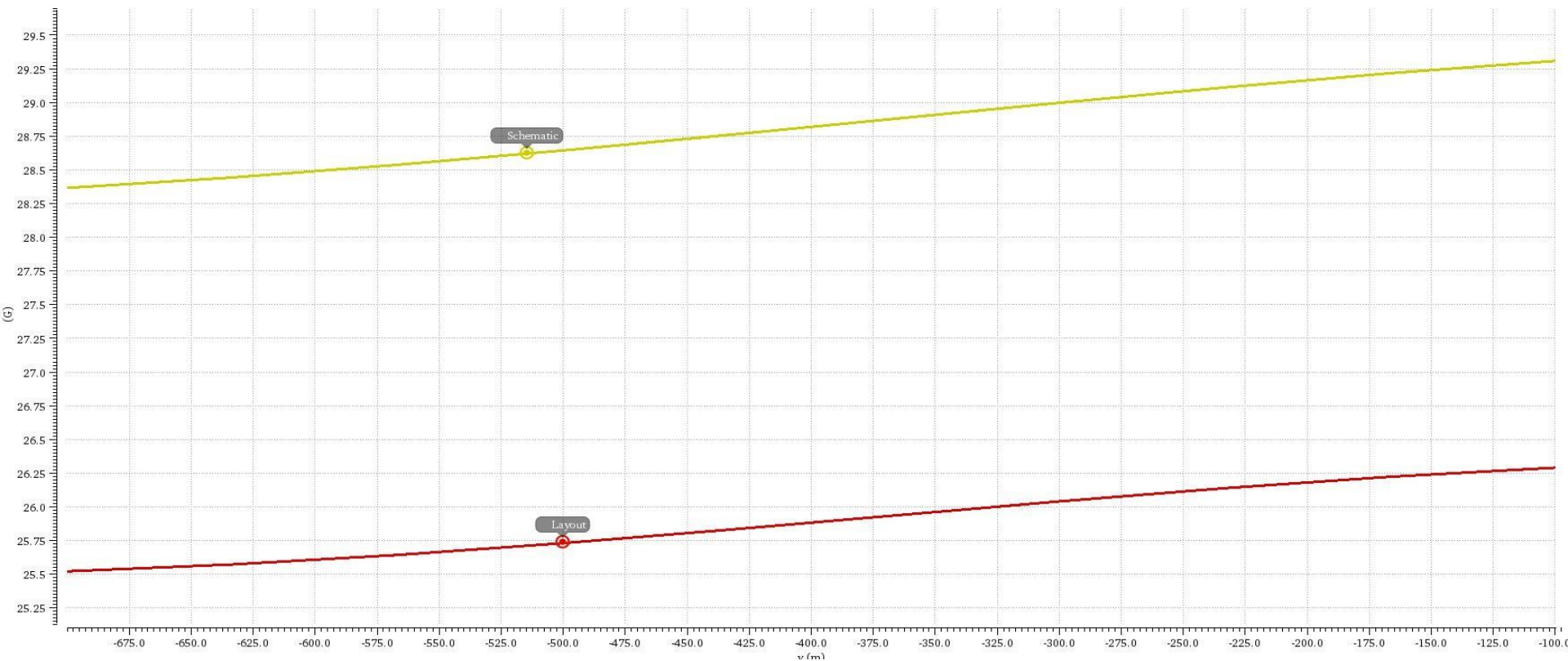




165 x 80



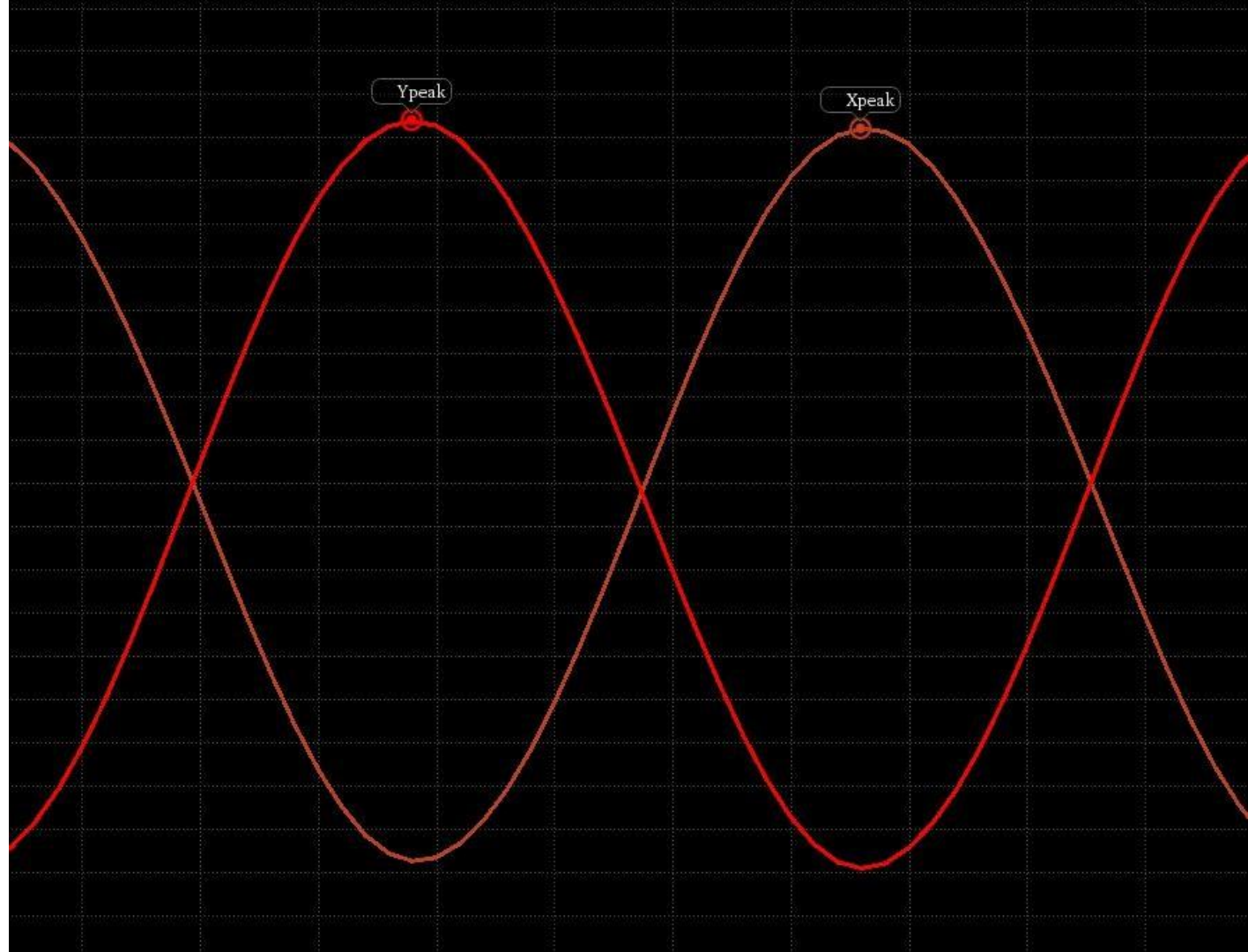
A comparison with the schematic and the layout proving the increase in capacitance resulting in drop in frequencies across all the levels.



Min freq 24.8014 GHz
Max freq 28.619 GHz

Vpp = 2.21 V to 1.35 V

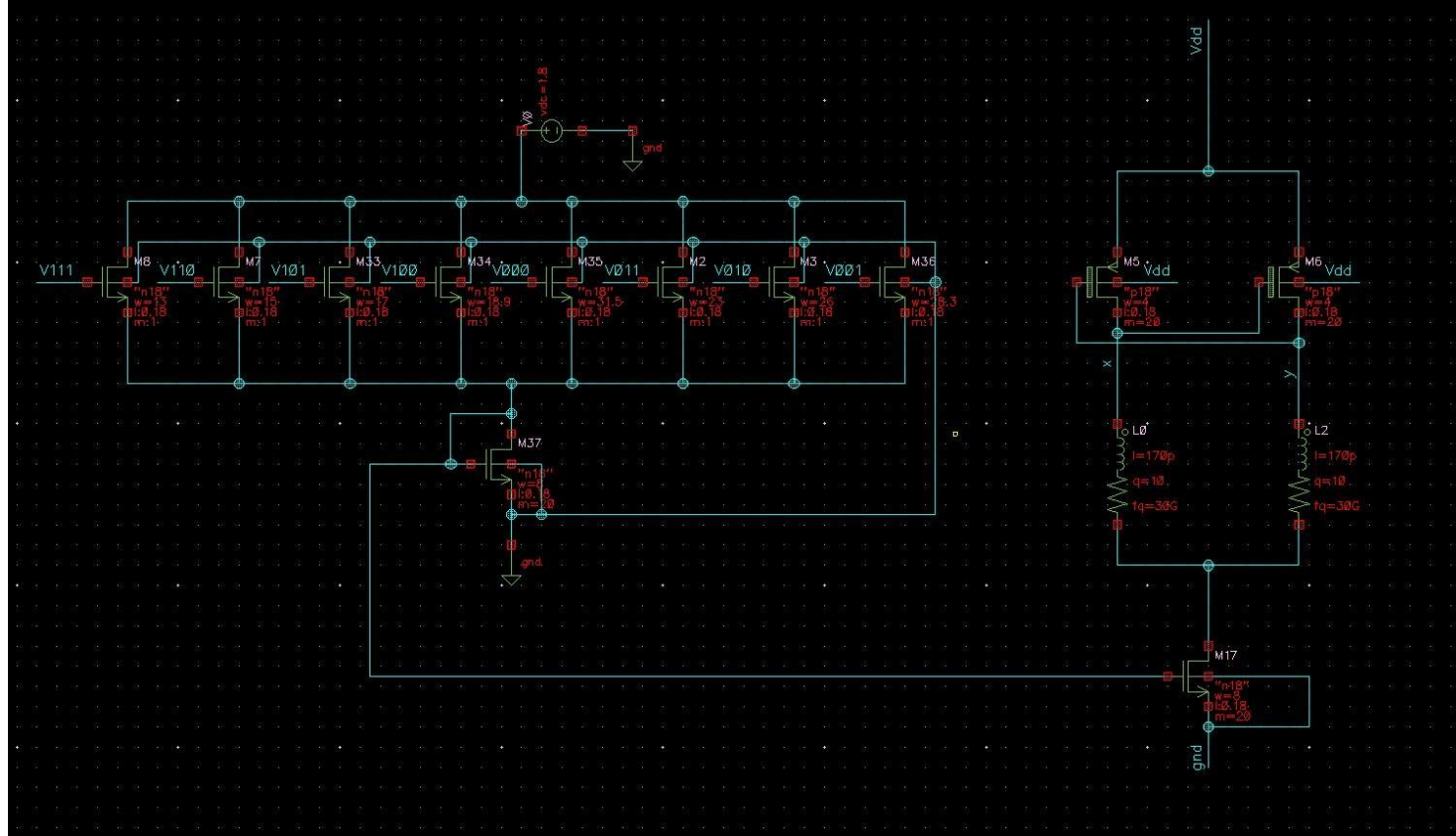
As X and Y are 180 degree
apart this was due to the
symmetry in the layout.



AMAR
KHATRI

Design of Current Taps for Voltage swing control

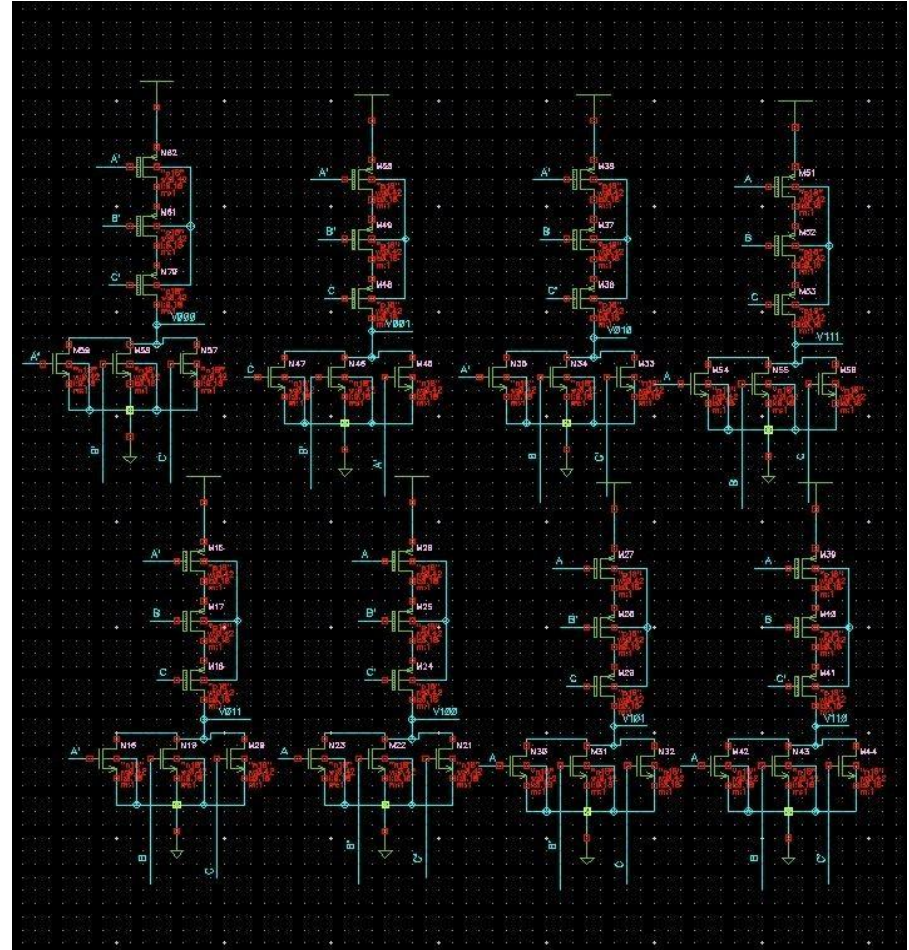
The design of the voltage divider ladder was changed to Current taps. The bottom nmos is designed in such a way that it can pass more current than the above nmos and the Current into the circuit in each Case is defined by the above active nmos.



Receiving Bit signal

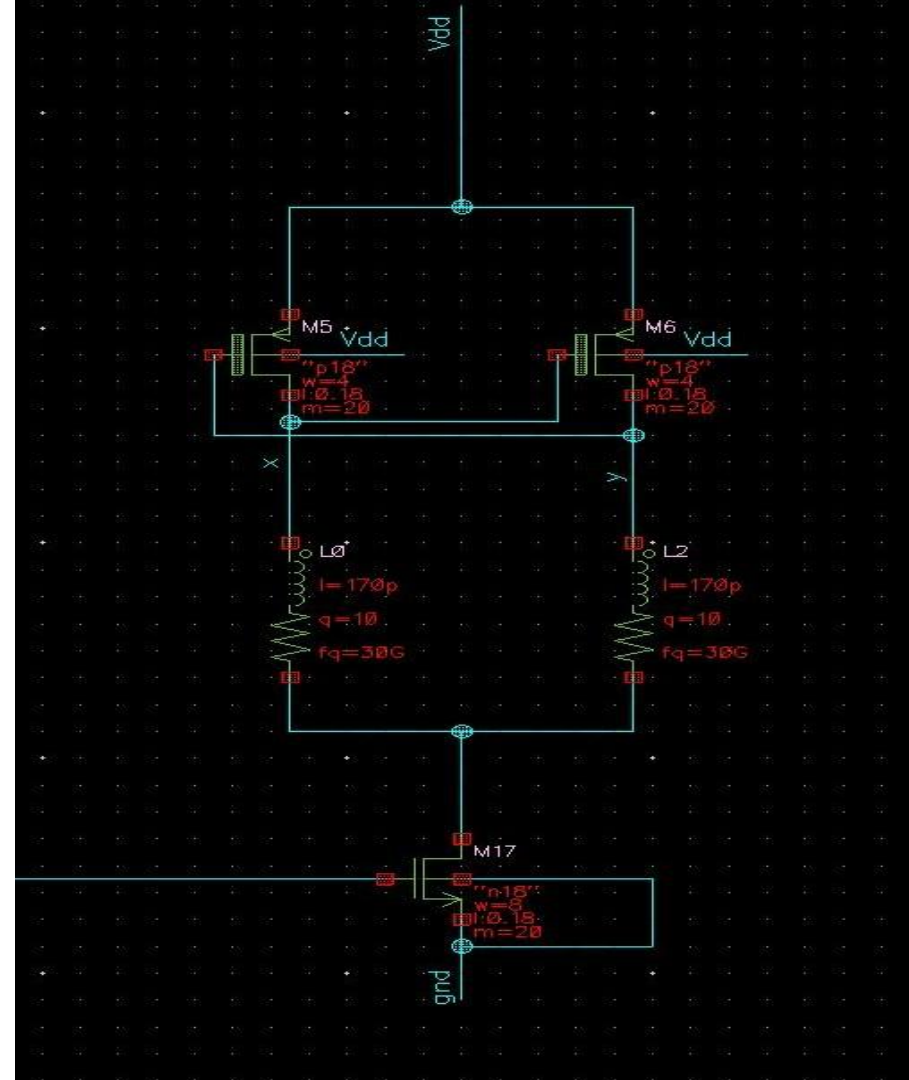
The A , B , C is the same control voltage that is used to switch ON and OFF the capacitor banks and it is also used to control the current.

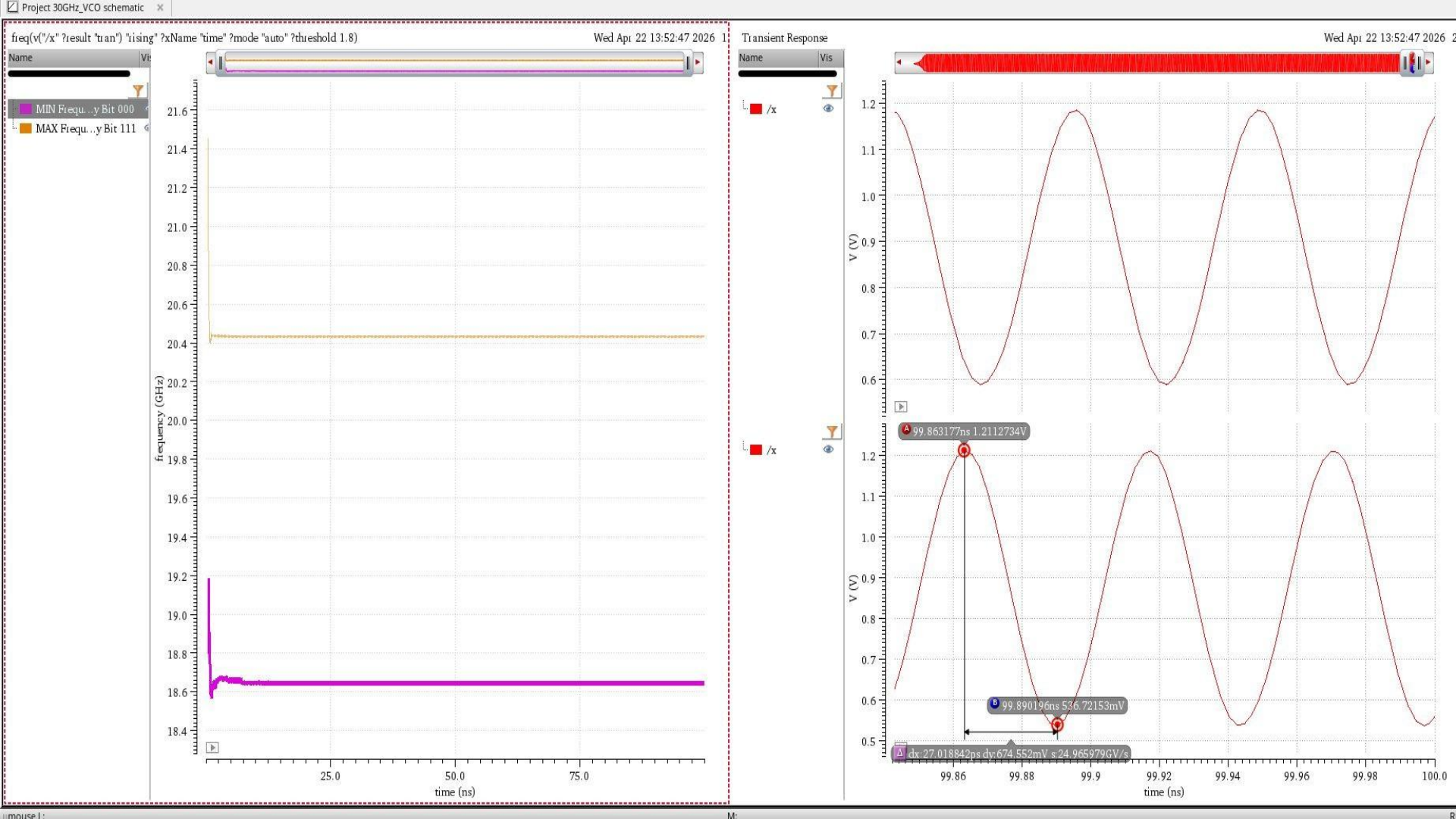
The current taps are then current mirrored to the tail current source nmos in the VCO



PMOS VCO

We tried different topologies we used cross coupled pmos transistor for generating negative resistance. But the problem we encountered was that the current driving strength of pmos is approx 3.5 times lesser than the nmos which caused reduce differential voltage swing. To compensate that we increased the width which led to the trade-off of Wide frequency tuning range.





Due to this increase in the parasitic capacitances of the the wide band specs changed drastically:

Nmos VCO

Tuning Range : 17.07 %

Max Freq = 32.1Ghz

Min Freq = 27.05 GHz

Pmos VCO:

Tuning Range : 9.94%

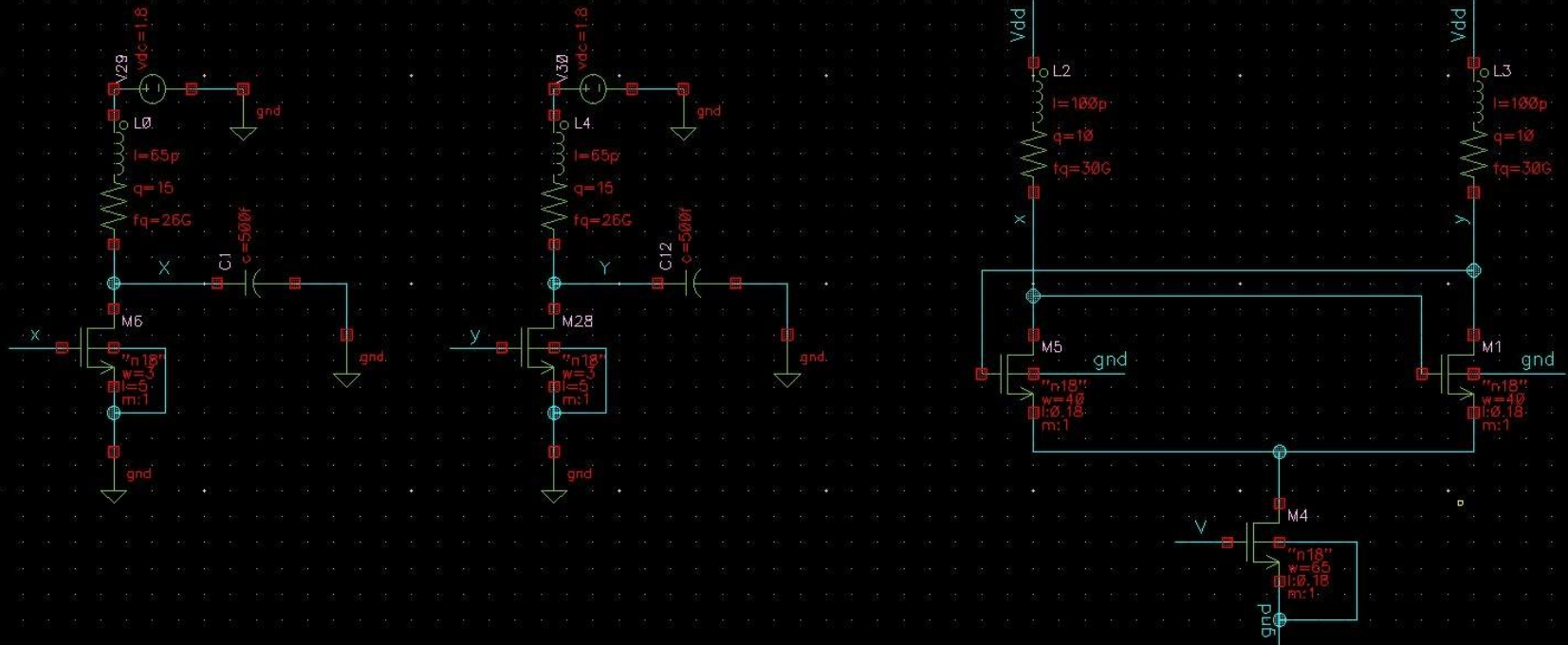
Max Freq = 20.43 GHz

Min Freq = 18.63 GHz

The Center Frequency dropped from 29.075 GHz to 19.5 GHz.

Due to these reason we did layout in the NMOS VCO as it adds another massive interconnect capacitance that would further decrease the frequency range and the wide frequency band.

Design of the Buffer



Buffer was needed to isolate the VCO from the

The resistance at the output node is very less as $\omega = 27\text{GHz}$

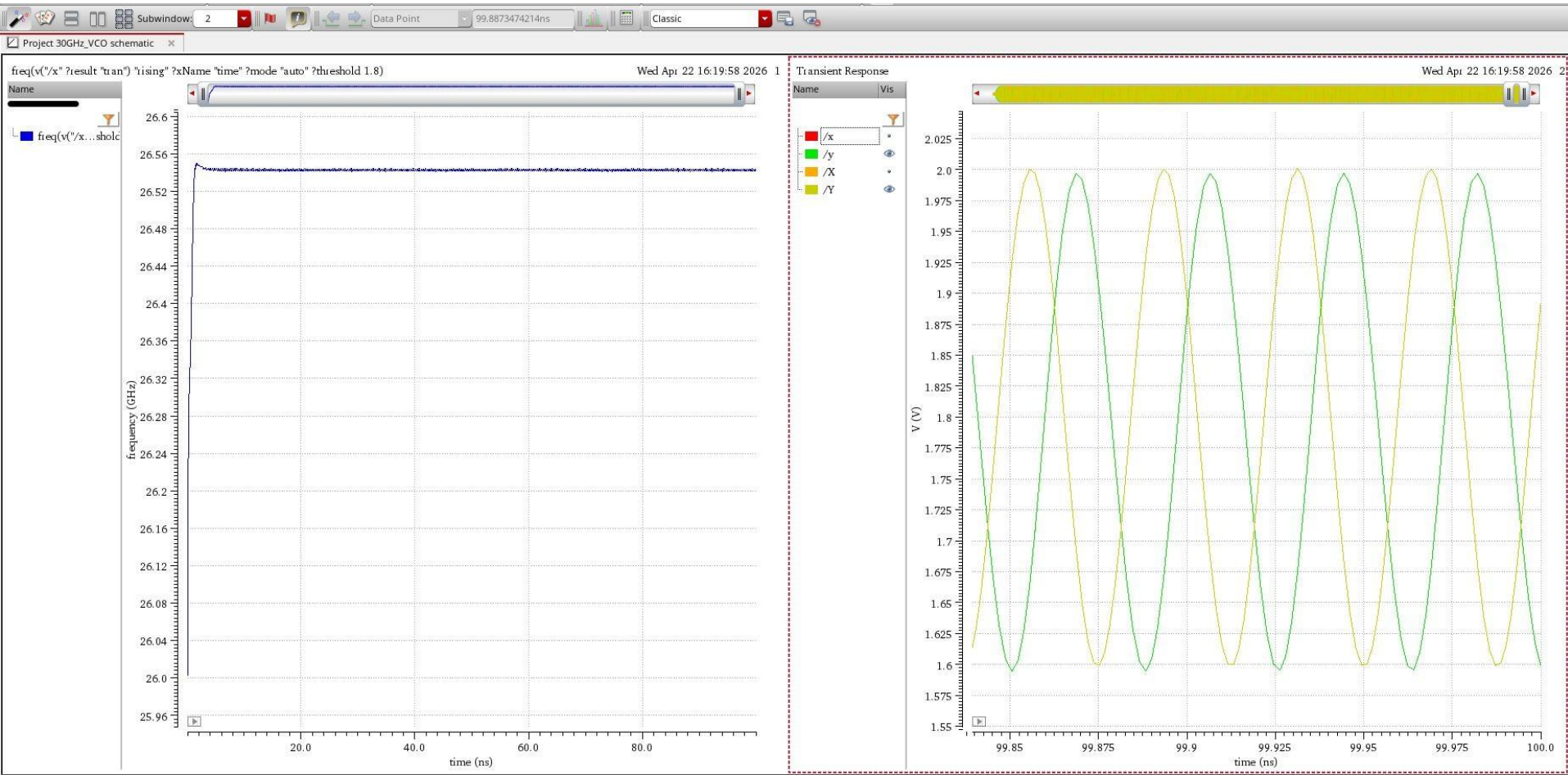
So X_c is approx 0.9 ohms

$X_l = 12$ ohms

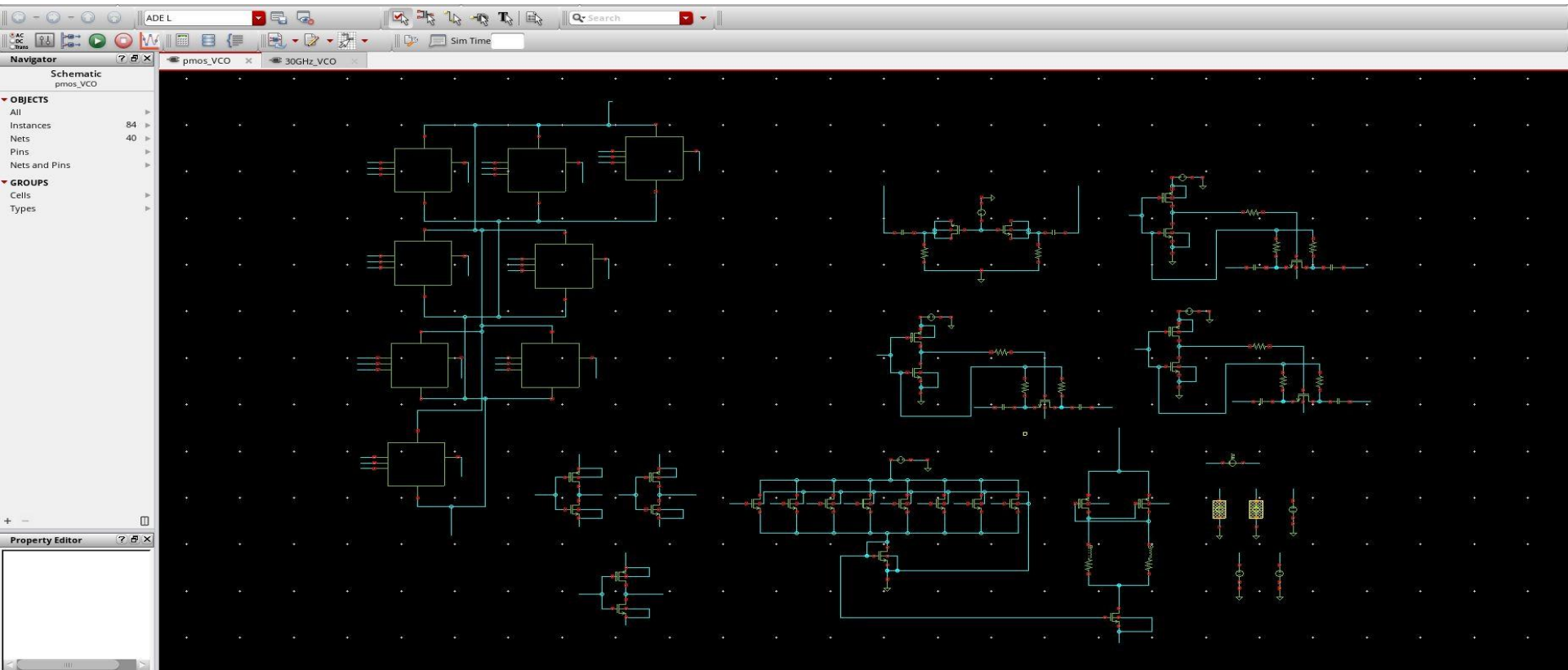
$R_p = 184$ ohms

So theoretically gain is coming very small but at resonance where $\omega_l = 1/\omega_c$ gain actually rises and becomes one. Still not clear why happens.

Results of Buffer



Pmos VCO



References

- 1) RF Microelectronics (2nd Edition) by Behzad Razavi

Research Paper reference:

- 1) Design of a 24-30 GHz Wide Tuning-Range VCO for 5G Millimeter Wave Communication by Zhijian Chen, Hong Chen, Bin Li, Yongxiang Li, Zhaohui Wu, Yuming Xu
- 2) A Wide-Band CMOS LC VCO With Linearized Coarse Tuning Characteristics by Jongsik Kim, Jaewook Shin, Seungsoo Kim, and Hyunchol Shin, Member, IEEE